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Effectiveness of a universal preschool-based program for emotional education in 3- to 5-year-old children with autism spectrum conditions

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Abstract

The "1,2,3,emoció!" is a universal preschool program of emotional education for children aged 3–5 years that has positive effect on emotional competence. The objective of this study was to evaluate the effectiveness of this program specifically in 3- to 5-year-old children with autism spectrum conditions. The effectiveness was evaluated during the first year of implementation (school year 2018–2019) in Barcelona. We conducted a cluster randomized community trial, with schools as units of randomization. The children's level of emotional competence was evaluated with the validated Emotional Competence Assessment Questionnaire (score range from 30 to 180). We designed a linear regression model for paired measures. The dependent variable was the final Emotional Competence Assessment Questionnaire score adjusted by the initial score, sex, and grade. The sample consisted of 49 children with autism spectrum conditions, of which 11 were girls. Children in the intervention group showed an increase from the initial score to the final score that was almost three times higher (16.1) than the increase observed in children in the comparison group (6.6). The difference between the two groups was statistically significant, with 8.9 points (95% confidence interval: 1.2–16.6) attributed to the intervention. We conclude that the "1,2,3,emoció!" universal program has the potential to be effective in children with autism spectrum conditions.

Lay abstract

There are numerous programs and interventions to improve social and emotional skills that expressly target children with autism spectrum conditions. However, there is less knowledge about the impact of universal (directed to all children) school programs of emotional education specifically on these children. In this context, the aim of our study was to evaluate the impact of the "1,2,3,emoció!" universal school program on them. The results show that children with autism spectrum conditions participating in the program improved their emotional competence more than the ones who did not participate. Therefore, we conclude that the "1,2,3,emoció!" universal program is effective in children with autism spectrum conditions. This promising outcome gives the opportunity to these children to benefit from group learning in an inclusive school setting, by complementing individual interventions with a universal program.

Keywords

autism spectrum disorders, cluster randomized trial, emotional competence, preschool children, universal intervention

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Introduction

Autism spectrum conditions (ASC) are characterized by persistent impairments in social communication and interaction, and restrictive and repetitive patterns of behavior, interests, or activities (American Psychiatric Association, 2013). Most children with ASC show difficulties in emotional regulation (Costescu et al., 2021) and in establishing positive social relationships (Kasari et al., 2016). In particular, inadequate emotional regulation is associated with anxious and depressive symptomatology, externalizing behaviors, and poorer quality of life, finally constituting an obstacle to social integration and good performance at school (Beck et al., 2020; Costescu et al., 2021).

Various programs to improve social skills and emotional regulation have been designed and implemented in children with ASC with good results (Lord et al., 2018). However, these good results, closely related to improving emotional competence, rarely generalize to children's natural environment (Kasari et al., 2016). Moreover, most of these programs start from 5 years of age (Beck et al., 2020), whereas the ideal age to work on the development of emotional competence is, at least in children of typical development, 3–5 years of age (Koglin & Petermann, 2011). Finally, the development of these emotional and social skills can also be enhanced through universal structured school programs (aimed at all children) of emotional and social education (Durlak et al., 2011). Yet, there is little evidence on the impact of universal school programs specifically on children with ASC.

The program "1,2,3,emoció!" (Bartroli et al., 2018) is a universal preschool program of emotional education for children aged 3–5 years developed by the Barcelona Public Health Agency. Its objective is to promote health and prevent risk behaviors through the improvement of emotional competence. This program was already shown to be effective in children without ASC (Pericas et al., 2022), significantly enhancing the emotional competences of participating children aged 3–5 years during the first year of implementation.

Therefore, the aim of this study was to evaluate its effectiveness in children with ASC during the 2018–2019 school year in Barcelona. Our hypothesis was that the program "1,2,3,emoció!" is effective in developing emotional competence in 3- to 5-year-old children with ASC.

Methods

Design

This study was a cluster randomized community trial, with schools as the unit of randomization.

Sample

In 2017, all schools in Barcelona were invited to participate in the evaluation of the effectiveness of the "1,2,3,emoció!" program. The research team randomly assigned schools that agreed to participate in either the intervention group (implementing the program) or the comparison group (not implementing the program). The OxMaR—Oxford Minimization and Randomization system was used for randomization. To ensure comparability between these two groups, schools were randomized stratifying by type (public or private/semi-private) and socioeconomic status of the school's neighborhood (low or high). The trial was conducted during the 2018–2019 school year. For this study, the study population was only that which met the following criteria: (1) children attending P3 (3-year-olds), P4 (4-year-olds), or P5 (5-year-olds); (2) enrolled in a school in the city of Barcelona participating in the study; and (3) diagnosed with ASC. Neither the IQ nor the developmental level of the children was assessed.

Intervention

The intervention consisted in the application of the "1,2,3,emoció!" program throughout a school year in P3, P4, and P5 classrooms. The program was applied to all children in the classes regardless of whether they had a diagnosis of ASC. This program is an adaptation of the Social and Emotional Aspects of Learning program developed by the Department of Education of the United Kingdom (Humphrey et al., 2008). The "1,2,3,emoció!" works on the five emotional skills described by Bisquerra (2003) in his emotional competence model: emotional awareness, emotional regulation, emotional autonomy, social competence, and life skills and well-being. The program includes a previous online training of 20 h for the teachers who implement it and has a didactic guide that includes, per course, 48 activities to carry out in the classroom and 6 activities to propose to families. It also has another complementary guide with 12 activities to be carried out in the school environment, outside the classroom (e.g. in the school canteen). The program consists of six thematic units: (1) Belonging, (2) Self-esteem, (3) Friendship, (4) Challenges, (5) Justice and bullying, and (6) Change, loss, and death.

Measures

The assessment instrument we used was the ECAQ (Emotional Competence Assessment Questionnaire), a questionnaire with strong validity and reliability evidence (Bartroli et al., 2022). It is answered by teachers, consists of 30 items, and uses a Likert-type scale with six categories (from 1 = Never to 6 = Always), with a total score ranging from 30 to 180. The higher the score, the better the emotional competence of the child. At both the beginning and end of the school year, tutors completed the questionnaire for each child (average time per child approximately 7 min).

Variables

The dependent variable was the final score of ECAQ. The main covariate was whether the child belonged to the

Table 1. Mean of initial ECAQ score and its increase, both for the intervention and comparison groups, and linear regression of final ECAQ score.

	N	Initial score		Increase in score		Linear regression of final score[a]		
		M	95% CI	M	95% CI	Coef.	95% CI	p value
Intervention group	34	65.8	59.6–72.0	16.1	11.7–20.5	8.9	1.2–16.6	0.024
Comparison group	15	76.5	66.3–86.6	6.6	1.5–11.7	Ref.	–	–

[a]Model adjusted for initial score of ECAQ, sex, and grade.

intervention group. Other covariates considered were the sex of the students and the preschool grade to which they belonged. Finally, exhaustivity was included to assess the quality of the implementation process, categorized into "high" (for children receiving four or more class activities per unit) and "low" (for those receiving less than four class activities per unit).

Analysis

Baseline equivalence between children in the intervention and comparison groups was determined by comparing proportions for each individual and contextual covariate using a chi-square test. A comparison of means was made using Student's t test for the children's initial level of emotional competence (initial score in the ECAQ) and for the increase in emotional competence according to whether or not they belonged to the intervention group. Subsequently, we designed a linear regression model for paired measures with final score of ECAQ as a dependent variable. We adjusted the model for the ECAQ initial score, sex, and grade of the children. We used STATA v.15 statistical program for data analysis.

Ethics

In order to guarantee the confidentiality and anonymity of the data, every teacher used a code to answer the ECAQ for each of the children they assessed. This code was created by the teacher. After the questionnaires were completed, the research team was the only one who had access to the results. Upon receiving the responses, the main researcher removed the codes from the database and assigned a new ID.

This study followed all the ethical guidelines for research with humans, and it was approved by the Ethics and Research Committee of Parc Salut Mar under the number 2019/8508/I.

Community involvement

Related to community involvement, children's teachers implemented the program in their classes, assessed children's emotional competence using the ECAQ, and promoted the family activities of the program. Families who wished to do so carried out the proposed family activities at home with their children.

Results

The sample consisted of 49 children, 11 of which were girls, from 25 different schools (average mean: 1.96 children per school). Thirty-four and 15 children were assigned, respectively, to the intervention and comparison groups. Among all of them, 38.8% were in P3, 26.5% in P4, and 34.7% in P5; 61.2% had low socioeconomic status, according to the school district's gross household income (Ajuntament de Barcelona, 2017); and 77.6% were attending public schools. No specific data on the race/ethnicity were recorded. Of the children receiving the intervention, exhaustivity was high for 85.3% of them.

Table 1 shows the mean initial ECAQ score and its increase, calculated as the difference between the mean final and initial scores, for both the intervention group and the comparison group. No significant difference was observed in the initial score between the two groups, and the score increased in both. However, the increase was significantly higher (almost three times) in the intervention group (16.1, 95% confidence interval (CI): 11.7–20.5) than in the comparison group (6.6, 95% CI: 1.5–11.7).

Table 1 also shows the results of the linear regression, using as dependent variable the final score of ECAQ adjusted for initial score, sex, and grade. It can be seen that belonging to the intervention group positively contributed to the difference between the final and initial ECAQ scores, with 8.9 points being attributed to the intervention (95% CI: 1.2–16.6).

Discussion

Our study shows that the "1,2,3,emoció!" universal program increases the emotional competence of children with ASC. Indeed, the increase in ECAQ score was higher in the intervention group than in the control group, and a difference of 8.9 points (95% CI: 1.2–16.6) was due to the intervention. This result is in line with the data obtained for children without ASC who previously participated in the "1,2,3,emoció!" program (Pericas et al., 2022).

Some studies show that specific programs are better than universal ones to increase emotional competence in

children with ASC (Kasari et al., 2016). In our study, we observed that in the application of a universal intervention in preschool, children with ASC also benefit. This is a major result, considering that the intervention is applied in the children's natural environment. Moreover, it is easier to implement universal programs than ASC-specific ones requiring specific resources that sometimes are less accessible. Finally, there might be children who are still not diagnosed with ASC who could benefit from the effect of a universal program. In any case, universal programs should not be considered as substitutes of the specific ones, but as good complements. Indeed, the group effect may play an important role, and some studies show how the participation of peers without ASC and the generalization of the social skills learned represent a positive support for children with ASC (Dean & Chang, 2021). Finally, the program's apparent success in improving children's emotional competence could also be attributed to the thorough and rigorous implementation of the program, as other authors have pointed out (Durlak & DuPre, 2008).

The most important limitation is that we use the teacher-reported outcomes in a trial where teachers are very involved in intervention. However, when teachers were asked "Did you feel that your work was under evaluation and that you should change it if a child got a low score in the test? In case you respond yes, in what way did you feel that your work was under evaluation?" and "Do you think that the fact of having to answer the test twice somehow influenced you in scoring the children? If yes, how do you think it influenced you?" 89% of them did not feel that their work was under evaluation and that they should change it in case a child got a low score, and 94% said that the fact of having to answer the test twice did not influence them in the task (Bartroli et al., 2022). Another limitation of this study is the small sample size. Nevertheless, it is noteworthy that, despite this, we found significant differences between the intervention and comparison groups. Finally, it is a limitation that the questionnaire is answered by teachers and not by children themselves. However, some studies suggest that early-childhood teachers, if trained and adequately supported, can make reliable judgments about children's social and emotional competence (Darling-Churchill & Lippman, 2016).

The main strength of this study is the analysis of the impact of a universal preschool intervention specifically on children with ASC. Another strength is its design, with a comparison group to control for the maturation bias (among other factors that could improve emotional competence), that could threaten the internal validity of the study. Moreover, the stratified randomization of schools to the intervention and comparison groups improved the homogeneity of the groups. Finally, the questionnaire used to assess emotional competence had already shown validity and reliability in 3- to 5-year-old children (Bartroli et al., 2022).

In conclusion, this study shows the potential positive effect of a universal preschool program for emotional education specifically on children with ASC in an inclusive school setting. Therefore, it seems that universal programs can be a complement to specific interventions to improve the emotional and social competence in children with ASC.

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